

Tutorial 1

Econometric Theory II

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- Fundamental Problem of Causal Inference. We cannot observe:

$$E[Y_i(0)|W_i = 1]$$

$$E[Y_i(1)|W_i = 0]$$

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- What if independence does not hold? (think of non random assignment of the treatment)

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- **Selection Bias** = *Difference in baseline outcome between treated and not treated units, prior to treatment*
- NOTE: for our purposes (DiD), selection bias *is not a problem* **IF** PTA holds. **The two groups can be different**, as long as their trends in outcome evolve in the same way (in absence of treatment).

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Let's see together where the Post-Treatment Bias comes from (whiteboard):

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- α_i : **unit fixed effect** (time-invariant unobserved heterogeneity)
- u_{it} : idiosyncratic error (time-varying shocks)

Why fixed effects? (endogeneity from omitted variables)

- If α_i is correlated with x_{it} , pooled OLS is biased:

$$\text{Cov}(x_{it}, \alpha_i) \neq 0 \Rightarrow \hat{\beta}_{\text{OLS}} \text{ biased}$$

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Typical examples:

- Wage $\rightarrow$ education, controlling for time-invariant ability
- Outcomes $\rightarrow$ policy exposure, controlling for permanent differences across places

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- α_i drops out $\Rightarrow$ FE can be estimated by OLS on demeaned variables
- $\hat{\beta}_{FE}$ uses **within** variation only

FE as least squares with unit dummies

Equivalent regression:

$$y_{it} = \beta x_{it} + \sum_{i=1}^{N-1} \gamma_i \mathbf{1}\{unit = i\} + u_{it}$$

- Include a dummy for each unit (minus one to avoid collinearity)

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Key implication: any time-invariant regressor (e.g. female, birthplace) is perfectly collinear with α_i and cannot be estimated.

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Interpretation: β is identified from deviations from both:

- each unit's average level, and
- each period's average level

Identification assumptions in FE models

For consistency of $\hat{\beta}_{FE}$ we need **strict exogeneity**:

$$\mathbb{E}[u_{it} \mid x_{i1}, \dots, x_{iT}, \alpha_i] = 0 \quad \forall t$$

Meaning:

- no feedback from past shocks to future regressors
- regressors are not chosen in response to unobserved shocks u_{it}

Violations:

- dynamic relationships (lagged dependent variables)
- policy reacts to outcomes (reverse causality)
- measurement error in x_{it}

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Let's see it together (whiteboard):

Main DiD Assumption

- **Parallel Trends Assumption (PTA)**: *in absence of treatment, the average outcomes for treated and control group **would** have followed parallel paths over time.*

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So what we care about is the **path of the outcomes AFTER the treatment takes place**, but in the hypothetical situation where nobody gets treated.

What if PTA doesn't hold?

Then the **counterfactual Δ trend would be $\neq 0$**
(Let's do it together on the whiteboard)

Dynamic DiD - Multiple Periods

$$Y_{it} = \alpha_i + \gamma_t + \sum_{k=-K, k \neq T-1}^K \theta_k \times W_{i,t+k} + \epsilon_{it}$$

or

$$Y_{it} = \alpha_i + \gamma_t + \sum_{\text{leads}=-K}^{T-1} \theta_{\text{leads}} \times W_{i,t+\text{leads}} + \sum_{\text{lags}=T}^K \theta_{\text{lags}} \times W_{i,t+\text{lags}} + \epsilon_{it}$$

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NOTE:

- 1 You have to drop one period to avoid collinearity.
- 2 Each coefficient in the dynamic specification tells the *differential change in Y relative to baseline!!!* Which in this case (and in general for convention) is in period $T - 1$.

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Remember that we are allowing for **trends** in the outcome!! What we want is just trends to be parallel and that they would continue to be that way without treatment. Then, both Ys are allowed to increase (or decrease) over time. $\hat{\theta}$ represents how much Y increase **more** for treated group compared to control.

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How should we think about the Leads?

- $\hat{\theta}_j = 0 \forall j < T - 1$ if trends are parallel before the treatment

Why to insert covariates in DiD?

- **Improve precision**

- Covariates that explain variation in the outcome reduce residual variance
- Leads to smaller standard errors and more precise estimates

- **Account for differential trends**

- If certain characteristics are associated with different outcome trends, interacting them with time can help satisfy the parallel trends assumption

- **But: Not required for identification (unless you rely on conditional PTA)**

- In standard DiD with fixed effects, identification relies on parallel trends
- Covariates mainly improve precision and credibility, not identification per se

What is a good covariate to insert?

- **Pre-treatment covariates**

- Variables determined before treatment occurs (like gender, age...)
- Avoids "bad control" problems
- Better add them while interacted with time dummies / time FE

- **Predictors of the outcome**

- Strong predictors increase efficiency

- **Avoid time varying controls!!!**

- It is very difficult to show that a time-varying control is not affected by the treatment...

What we will see now in the R script, which was missing here

- Balance tests of the covariates
- Ways to assess PTA
 - Graphical analysis
 - Joint Significance test of the leads
 - Placebo tests
 - Pre-trends tests
- How to add covariates
- Randomization inference